STM32F746 - Device Configuration

APB1 = 108 MHz

TIM2
Independent
TimeBase = TIME_BASE-2
Counter Up
global interrupt: CNT_EN

TIME_BASE = 4320

APB2 = 216MHz

ITR1 (TRGO)

TIM1
Gated Mode
TimeBase = TIME_BASE-1
Counter Center Aligned Mode 2

TIME_BASE = 4320
DEAT_TIME = 90

PWM_CH1
PWM_CH1N
PWM_CH2
PWM_CH2N
PWM_CH3
PWM_CH3N

Main task
launched via software
via
TIM2 global Interrupt

global_state_machine_process();

TRGO(2)

ADC1 (independent mode)
single mode - 12Bits
synch TRGO(2) from TIM1
scan mode - 4 channels
DMA - countinuos request

DMA - circular buffer

ADC2 (independent mode)
single mode - 12Bits
synch TRGO(2) from TIM1
scan mode - 4 channels
DMA - countinuos request

DMA - circular buffer

ADC3 (independent mode)
single mode - 12Bits
synch TRGO(2) from TIM1
scan mode - 6 channels
DMA - countinuos request

DMA - circular buffer

ADC_CLK = 108 MHz

CAN-BUS
Prescaler = 6 (54 MHz)
Quanta seg. 1 = 15
Quanta seg. 2 = 2
Boute rate = 500 kbit/s

IWD
Prescaler = 4 (32 kHz)
Window = 256
Count down reload = 6

OCD phase u

OCD phase v

OCD phase w

PWM Enable

BRAKE Enable

EXT INT

EXT INT

EXT INT

TIM1 BRIN

TIM1 BRIN2

TIM4
PWM - Fan